Kumud Jalori

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.44
B M Convent Hr Sec School	Class XII (CBSE)	2024	94.80%
B M Convent Hr Sec School	Class X (CBSE)	2022	97.80%

SKILLS

Programming Languages: C, C++, Python, SQL

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn

AI & GenAI: Generative AI(LLMs), Prompt Engineering, AI Productivity Tools

Tools: MS Excel, MS PowerPoint, MS Word, Power BI, Tableau, VS Code, Notion, Canva, Figma

Soft Skills: Communication, Problem Solving, Team Management, Critical Thinking, Leadership, Adaptability

PROJECTS

Local RAG Pipeline Implementation

- **Built a complete RAG pipeline from scratch** with PDF parsing, text preprocessing, sentence-based chunking, and structured knowledge-unit construction.
- **Generated GPU-accelerated embeddings using MPNet/OpenAI models** with batched encoding, vector storage, and efficient cosine/dot-product similarity search.
- **Developed semantic retrieval logic** with top-k ranking, ANN indexing concepts (IVF-Flat, HNSW), similarity scoring, and precise page-level traceability.
- **Implemented context assembly** by selecting, ordering, and merging the highest-relevance chunks for grounded generation.
- **Performed local LLM inference** by injecting retrieved context into generation workflows to ensure factual, hallucination-resistant outputs.

Stock Price Forecasting (Time Series Modeling using LSTM)

- **Built a time series prediction model using LSTM** (Long Short-Term Memory) networks to forecast stock prices from historical data.
- Developed a pipeline consisting of **feature scaling (MinMaxScaler)** and a time step creation of to prepare historical stock data for the input to the **RNN**.
- Trained the model using **Adam Optimizer** with a time window of 60 days to predict the single next opening price of the stock.
- Achieved a **low RMSE of 15.04 and MAE of 13.48**, though a little off with the real-time values, it accurately predicted the rise and fall of the stock.

Neural Networks Implementation from First Principles

- **Built a Multi-Layer Perceptron** from the ground up using NumPy, including Forward and Backward Propagation without using Keras.
- Developed custom functions for essential neural network components, including the **Softmax** output function and **Categorical Cross Entropy loss function**.
- Integrated the **Adam Optimizer** to manage the learning rates and update weights and biases.
- Achieved high-performance results on the dataset with a final **Categorical Cross-Entropy Loss of 0.05**.

POSITIONS OF RESPONSIBILITY

Inbound Head | International Relations Cell, IIT Hyderabad

- Supported international students from **Japan, Africa, and partner universities** through campus integration.

- Leading a team of **20+** coordinators, managing international student onboarding, orientation, exchange activities, and cultural engagement initiatives while coordinating with faculty and university stakeholders.

Internship Coordinator | Office of Career Services (OCS), IIT Hyderabad

- Coordinate with the Office of Career Services, recruiters and companies to support campus internship hiring initiatives.

Consulting Coordinator | Finance & Consulting Club, IIT Hyderabad

- **Co-authored** the FCC IITH Product Management & Consulting **Prepbook**.
- **Conducted PM workshops** for external participants and organized **CaseKnights** sessions for IITH students.
- **Organized institute-level case competitions** featuring live industry cases.

Events & PRO Coordinator | Tinkerers' Lab, IIT Hyderabad

- **Led sponsorship outreach** and industry collaborations for IIT Hyderabad's flagship technical fest, partnering with organizations including Cubelelo and supporting end-to-end event execution.
- **Managed digital outreach** and post-event documentation to support event visibility and execution.

EXTRACURRICULAR

- **Community Service:** Volunteered with the **National Service Scheme (NSS)** in social outreach and community service initiatives.
- **Creative Design:** Designed presentation decks, posters, and visual assets; actively pursue illustration and paper crafts.
- **Counselling Support:** Mentored 27 first-year undergraduate students as a **Sunshine UG Mentor**, providing academic guidance, peer support, and personalized mentorship to facilitate a smooth transition to campus life and university culture.